

Reminder: Your computer must be able to support unicode characters, especially emojis, for this program.

There are two species of plants involved in this simulation. The first species is Flower, and the second species is Tree. Each plant also has one of the following conditions: Seed, Small, Adult, Dead, Illness, FFlower, SFlower, which will be elaborated upon later. Each plant starts off at age 0. With each increasing step, the age of the plant increases by 1 as long as the condition of the plant is not Dead. If the plant dies(The condition is set to Dead), then the age is reset to 0.

Here are the corresponding graphics for plants of different species and conditions:

Condition	Flower	Tree
Dead		
Seed		
Small		
Adult		
Illness		
FFlower		Tree cannot be set to this
SFlower		Tree cannot be set to this

Initially, the garden starts off as a garden full of plants of randomized species, all age 0, and all set with the condition of Dead. If the garden is initialized as non-empty, all the plants should be represented by . With every step, while the plant's condition is dead, there is a $\frac{1}{4}$ possibility of a new plant being born to replace the old one, with a $\frac{1}{2}$ chance of it being a Tree and a $\frac{1}{2}$ chance of it being a Flower. If the plant stays dead, its age stays at 0 with every step. If a new plant is born, the new plant's condition will be Seed() , and the age will be set at 0, and the plant's age will increase with every step from here on as long as the plant stays alive.

With every step, there is a $\frac{1}{20}$ chance of the plant, regardless of its age, species, and condition, dying suddenly. If the plant is suddenly killed this way, its condition is set to Dead and its age set to 0. Initially, I went for a much higher possibility of death, but that caused a lot of plants to die early, which decreases the amount of variety in a garden as many plants don't reach adulthood, so I set the chance of a plant dying randomly to be very low.

From ages 0-9, the plant, regardless of species, will have its condition as Seed, unless it suddenly dies. From ages 10-14, the plant, regardless of species, will have its condition as Small, unless it suddenly dies.

From ages 15 and onwards, the plant can have several different conditions apart from the default condition, Adult, which it is initially set to right after turning 15.

Once the plant reaches the age of adulthood(ages 15 and beyond), for every step, it has a 1/10 chance of catching an illness, which is essentially setting the condition of the plant to Illness. Once a plant has the condition Illness, at each step, it has varied chances of being healed(condition set to Adult), staying ill(condition stays as Illness), and dying(condition set to Dead). If the plant is of ages 15-19, it has a 8/15 chance to heal, 1/15 chance to die, and 6/15 chance to stay ill. If the plant is of ages 20-34, it has a 8/20 chance to heal, 6/21 chance to die, and 6/20 chance to stay ill. If the plant is older than 34, it has a 8/25 chance to heal, 11/26 chance to die, and 6/25 chance to stay ill.

Thus, it's shown that with increasing age, an ill plant is increasingly more likely to die. This design fits with natural plants, with younger, stronger plants being more able to weather illnesses, while older ones have less resistance. The varied possibilities of how plants respond to illness also create more possible combinations of plant behaviors in the garden. I want to include this because it's natural for plants to become sickly for various reasons in the natural world, and the healing process for them produces a lot of random, organic behaviors. For the plant illnesses, I initially planned for plants of all ages to have equal possibilities of healing, dying, and staying sick, but then I realized in order to more closely mimic nature and to produce more varied behaviors, I might want to adjust the possibilities based on age.

Once a Flower reaches adulthood, as long as it is not ill or dead, with each step, there is a $\frac{1}{4}$ chance of it blooming(Having its condition set to FFlower or SFlower). There is a $\frac{1}{2}$ chance of its condition being set to FFlower and a $\frac{1}{2}$ chance of its condition being set to SFlower. FFlower and SFlower are two different flower blooms that an adult Flower might produce. However, once a Flower blooms, with every step, there is a $\frac{1}{4}$ chance of the flower closing, and the condition of the plant will be reverted back to Adult. Otherwise, the condition of the plant will remain FFlower or SFlower. Do note that regardless of whether a Flower is blooming or not, it could always catch an illness at every step, as it is an adult plant. I initially only had one flower variety(FFlower), but then I decided to add another for a bit more organic variation, as natural plants do produce different blooms depending on circumstances, which adds some extra variability to the garden.

Once a plant reaches the age of 50, it dies and its condition is set to Dead, and its age set to 0. Afterall, no immortal plants exist in nature.

Overall, my garden design has biodiversity(2 different species, and one species has two different flower blooms), representation of different ages(Seed, Small(young age), Adult, Dead), and random organic occurrences at different times and probabilities such as germination, becoming ill and healing from illness, blooming and closing, and dying suddenly. Therefore, the resulting behavior of my simulation is interesting and organic.